“From Behavioral Segments to Persona” was demonstrated by using the dataset (/kaggle/input/lifestyle-dataset/genz_digital_lifestyle_tech_habits.csv). The dataset contained 500 rows representing real-world digital lifestyle of GenZ. The dataset was synthetic and created for research and analysis purposes. The variables used for this project were User_ID, Age, Gender, Daily_Screen_Time_hrs, Preferred_Device, Social_Media_Usage_hrs, Gaming_Activity, Tech_Savviness, Streaming_Subscriptions and Privacy_Awareness_Level. All analyses were carried out in Python.

There were no missing values, no strong outliers, and no marked skewness. After checking the distribution of the variables, two standard diagnostics (Kaiser-Meyer-Olkin and Bartlett’s test) were performed as an initial step for the classical Factor analysis. Overall KMO was 0.282 and Bartlett’s p-value was 3.241477897826943e-65. (Even after dropping the variables according to KMO per variable <0.48, the overall KMO was only 0.502). Although Bartlett’s test was significant and variables seemed to be correlated, KMO was very low. This indicated the correlation structure that was not suitable for the factor analysis.

The dataset contains both numerical and categorical (ordinal & nominal) variables. So, Factor Analysis of Mixed Data (FAMD) was the most appropriate option to deal with the mixed data, reducing the variables into a smaller set of continuous dimensions while respecting the mixed nature of the data.

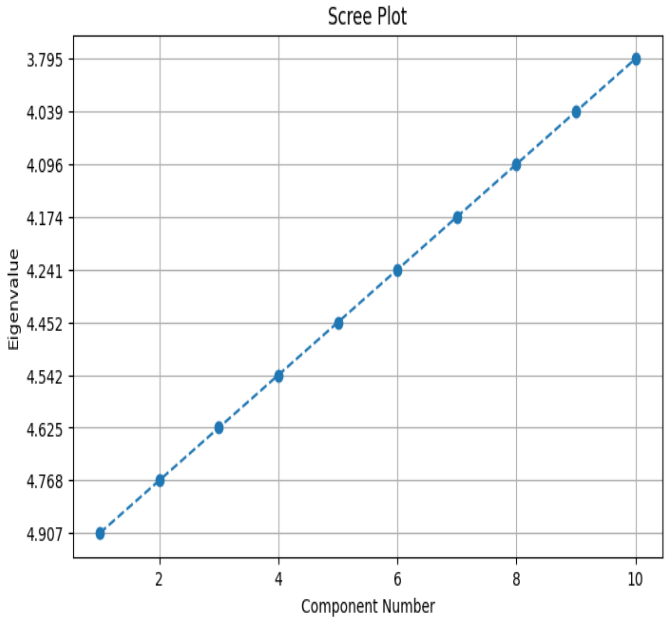


Figure (1) Scree plot with 10 components

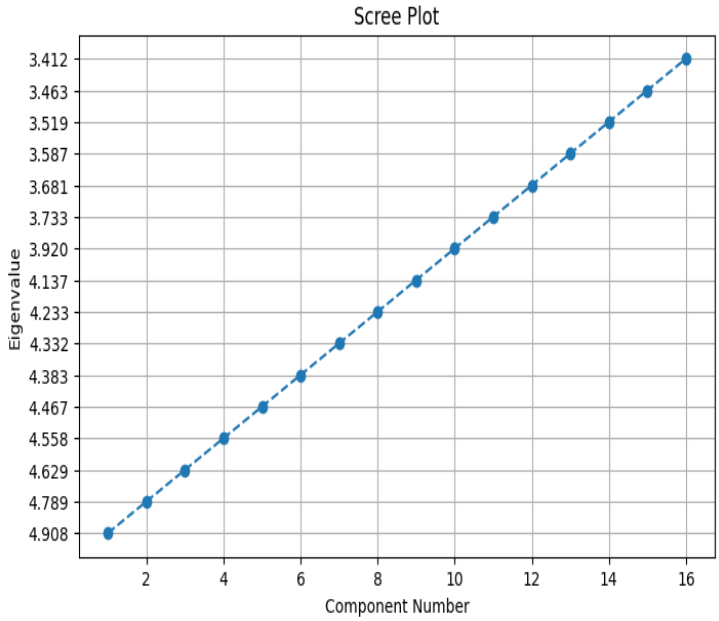


Figure (2) Scree plot with 16 components

I started fitting the FAMD with (5) components. The eigenvalues range from 4.34 to 4.85, which reflects that the structure was relatively spread across these first few dimensions and there was no clear elbow within the first (5) components. So, FAMD was run with more components i.e., (10) components (Figure – 1) and up to the maximum (16) components (Figure – 2). The scree plot did not show a clear elbow:

eigenvalues decreased almost linearly, indicating that the variance is spread across several dimensions rather than dominated by one or two. Subsequently, the performance of k-means clustering (silhouette score) was compared across (5) components. I retained 2 components that provided the highest silhouette score (0.36) with 3 clusters for subsequent analyses.

Interpretation of Dimensions

Numerical Variables

Dimension 1: It is primarily about Streaming Subscriptions.

- **Streaming Subscriptions (0.168):** This is the strongest positive link. Dimension 1 is mostly defined by people who have more paid services.
- **Age (0.059) & Social Media Usage (0.035):** These have a very weak positive relationship.
- **Daily Screen Time (-0.038):** Interestingly, it moves slightly towards the negative end.
- **Interpretation:** This suggests Dimension 1 represents "Quality/Paid Content" rather than just "Time Spent."

Dimension 2: Despite very low coefficients, it captures the contrast between Social Media Usage and Age.

- **Social Media Usage (0.081):** This is the strongest positive driver here.
- **Age (-0.065), Daily Screen Time (-0.056) & Streaming (-0.062):** These move in the opposite direction.
- **Interpretation:** Dimension 2 seems to capture a "Youthful Social" trend—as social media use goes up, age and traditional streaming subscriptions tend to go down (or vice versa).

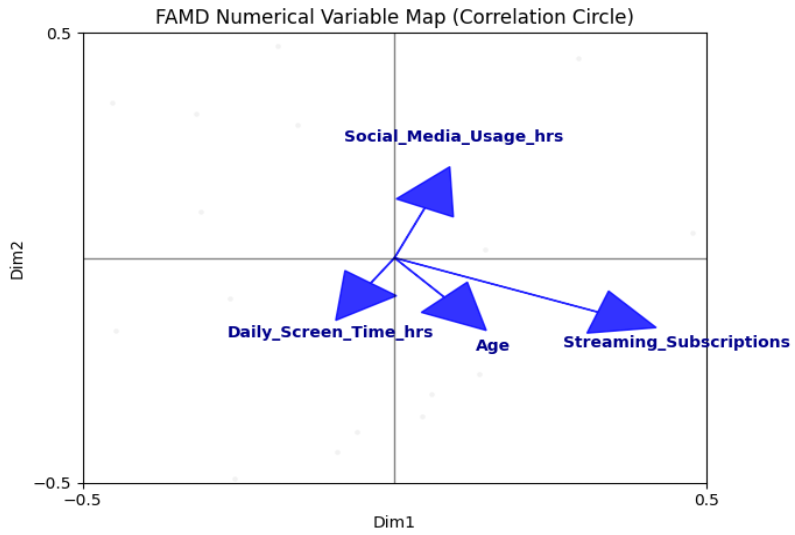


Figure (3) FAMD Numerical Variable Map

Categorical Variables

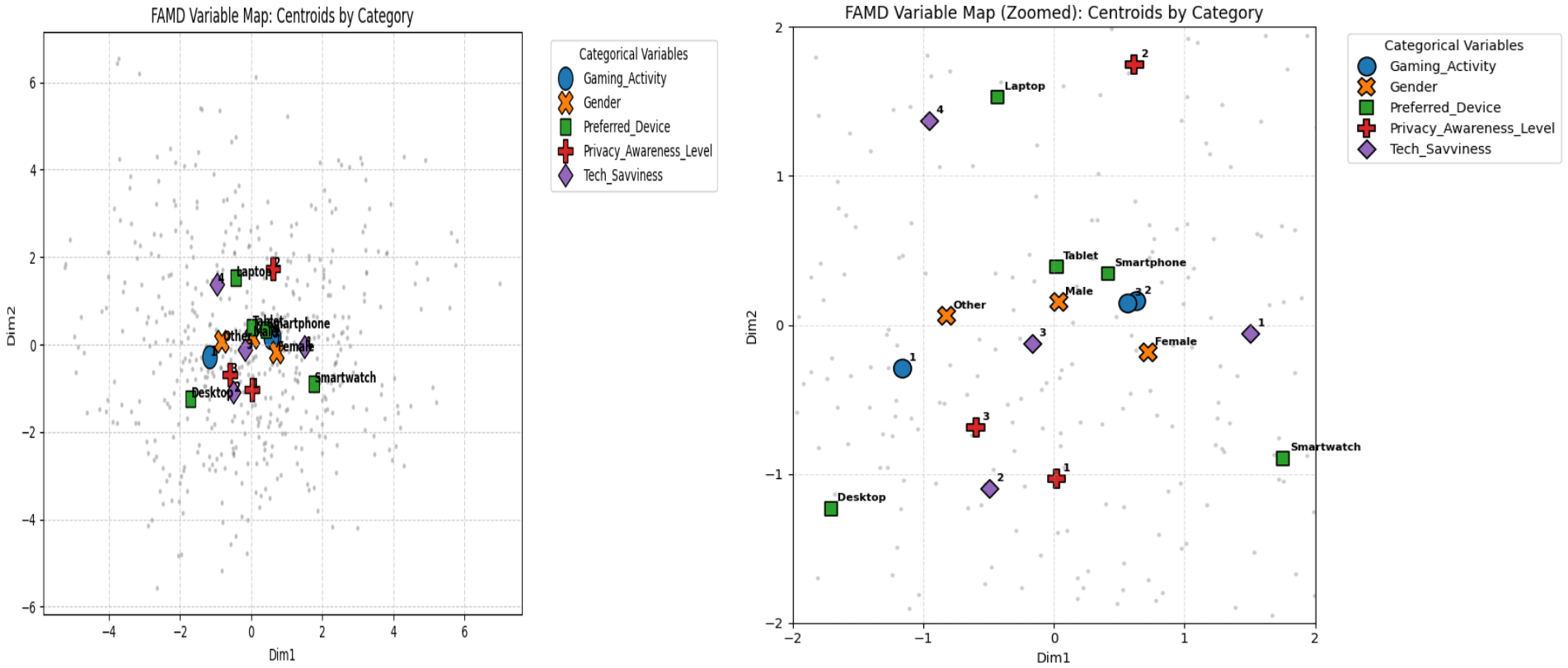


Figure (4) FAMD Variable Map: Centroids by Category

Dimension 1: Non-gamers, people with high privacy awareness, people who have significant practical knowledge of tech, professional style hardware users (desktop/laptop) and 'other' gender type have a negative influence, whereas gamers, people with beginner tech savviness level, tech users with a focus on mobility (e.g., smartphone, smartwatch), people with less privacy awareness and females move toward the positive end.

More specifically,

- **Gaming Activity:** Dimension (1) primarily distinguishes between "Gamers" and "Non-Gamers". High negative value (-1.158) is associated with those who never play; and positive values (0.633 and 0.566) are associated with those who play (occasional or frequent).

- **Tech Savviness:** Dimension (1) also seems to differentiate the “Beginner Level” from the upper levels regarding tech shrewdness. “Beginner” has a strong positive coordinate (1.507); and “Intermediate”, “Advanced”, and “Expert” have negative coordinates (-0.490, -0.161, -0.951 respectively).

- **Privacy Awareness Level:** In Dimension (1), “Medium” level of privacy awareness moves toward the positive side (0.615), whereas “High” level moves in the opposite direction (-0.596). So, Dimension (1) focuses on differentiation of high level of privacy awareness from the rest.

- **Preferred Device:** Smartwatch users and Desktop users are on the far positive (1.756) and negative (-1.705) sides of Dimension (1). Meanwhile, laptop and smartphone usage are relatively closer to center with (-0.430) and (0.417) respectively; and the closet is the tablet users (0.020).

- **Gender:** “Other” gender type has a strong influence (-0.821) on the negative side of Dimension (1), while “Female” leans toward positive side with (0.721). The influence of “Male” is almost negligible.

Dimension 2: The positive side represents gamers, people with expert tech savviness, ‘other’ gender type and mobile tech users (laptop and smartphone), while females, people with less tech savviness, non-gamers and desktop or smartwatch users are on the negative side.

- **Gaming Activity:** Since both "Occasional" and "Frequent" are on the positive side (0.159 & 0.144) while "Never" is on the negative side (-0.293), Dimension (2) does not strongly differentiate between the levels of gaming activity.

- **Tech Savviness:** Only “Expert” level (1.367) moves towards the positive side of Dimension (2) whereas the lower levels (“Beginner”, “Intermediate” and “Advanced”) are on the negative end with (-0.060, -1.100, -0.127) respectively.

- **Privacy Awareness Level:** Dimension (2) is unlikely to capture a linear relationship since Medium Level has a very high value (1.74), but Low Level (-1.03) and High Level (-0.68) are both negative.

- **Preferred Device:** Laptop users (1.52) are the extreme positive; and Desktop users (-1.23) and Smartwatch users (-0.89) are on the negative side. Tablet and smartphone users are relatively closer to the zero with (0.393) and (0.343) respectively.

- **Gender:** Females are on the negative side of Dimension (2) and the other two are on the positive side with weak influences.

Interpretation of Clusters

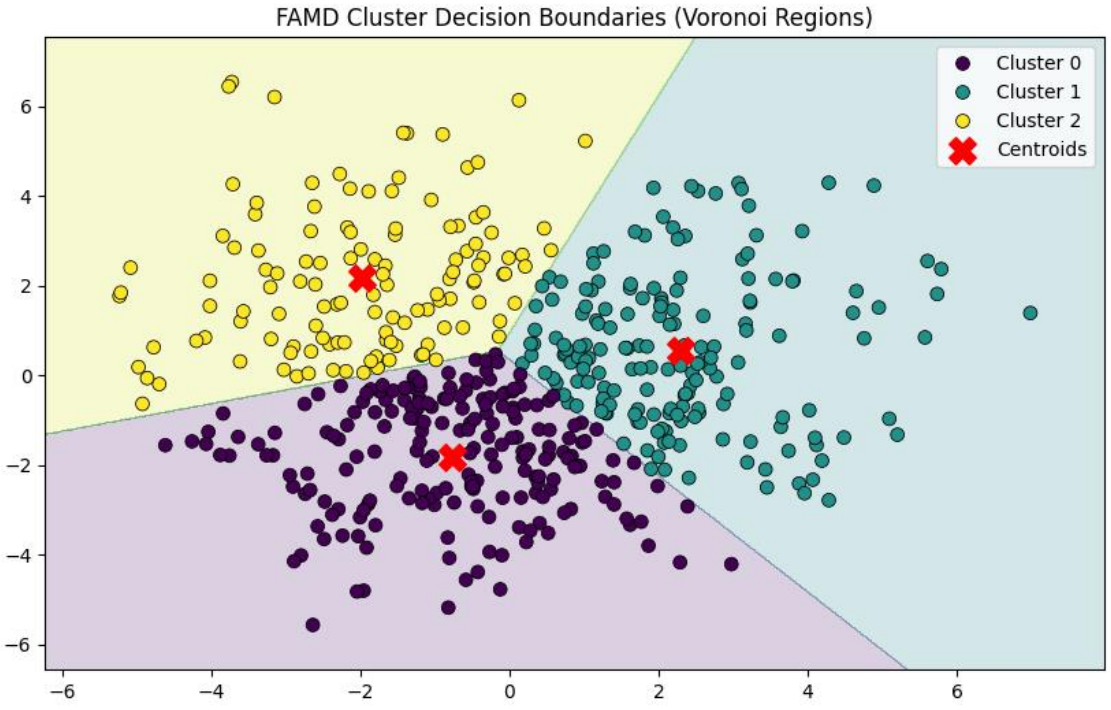


Figure (5) FAMD Cluster Decision Boundaries

Cluster 0: Desktop Traditionalist or Stationary Surfer

- **Age:** The middle-aged group (median ~ 24, and most concentrated around 24).
- **Gender:** A relatively even split between Female and Other gender group.
- **Gaming & Tech Astuteness:** They are not big gamers, but they lean more toward 'Occasional' than 'Frequent'. The highest proportion of 'Intermediate' level tech knowledge.
- **Behavior:** With the highest proportion of the hight level of privacy concern, the cluster spends time on screen more than the other two do, likely engaging in paid subscription than social media
- **Device:** This group is heavily 'Desktop' focused.

Cluster 1: The young tech-forward users

- **Age:** The youngest demographic (median ~22)
- **Gender:** The highest proportion of females.
- **Gaming & Tech Astuteness:** Highest level of gaming activity, and lowest tech savviness or probably a specific type of hobbyist.
- **Behavior:** Interestingly, they have the highest Streaming Subscriptions and social media usage with a significant level of privacy awareness.

- **Device:** Highest usage of smartwatch and smartphone, with a focus on portable devices such as tablets and laptop.

Cluster 2: The old, tech savvy with minimal digital footprint

- **Age:** The oldest group on average (median ~25), but with a very wide spread.
- **Gender: '**Other' gender type is the largest.
- **Gaming & Tech Astuteness:** Lowest Gaming Activity and highest proportion of 'Expert' level tech savviness, also with an overall proportion of tech savvy
- **Behavior:** The 'Privacy-Focused' cluster exhibits the minimal digital footprint, with the lowest overall duration spent on social media and streaming services.
- **Device:** Highest usage of laptops and tablets, with the least interest in using smartphones and smartwatches.

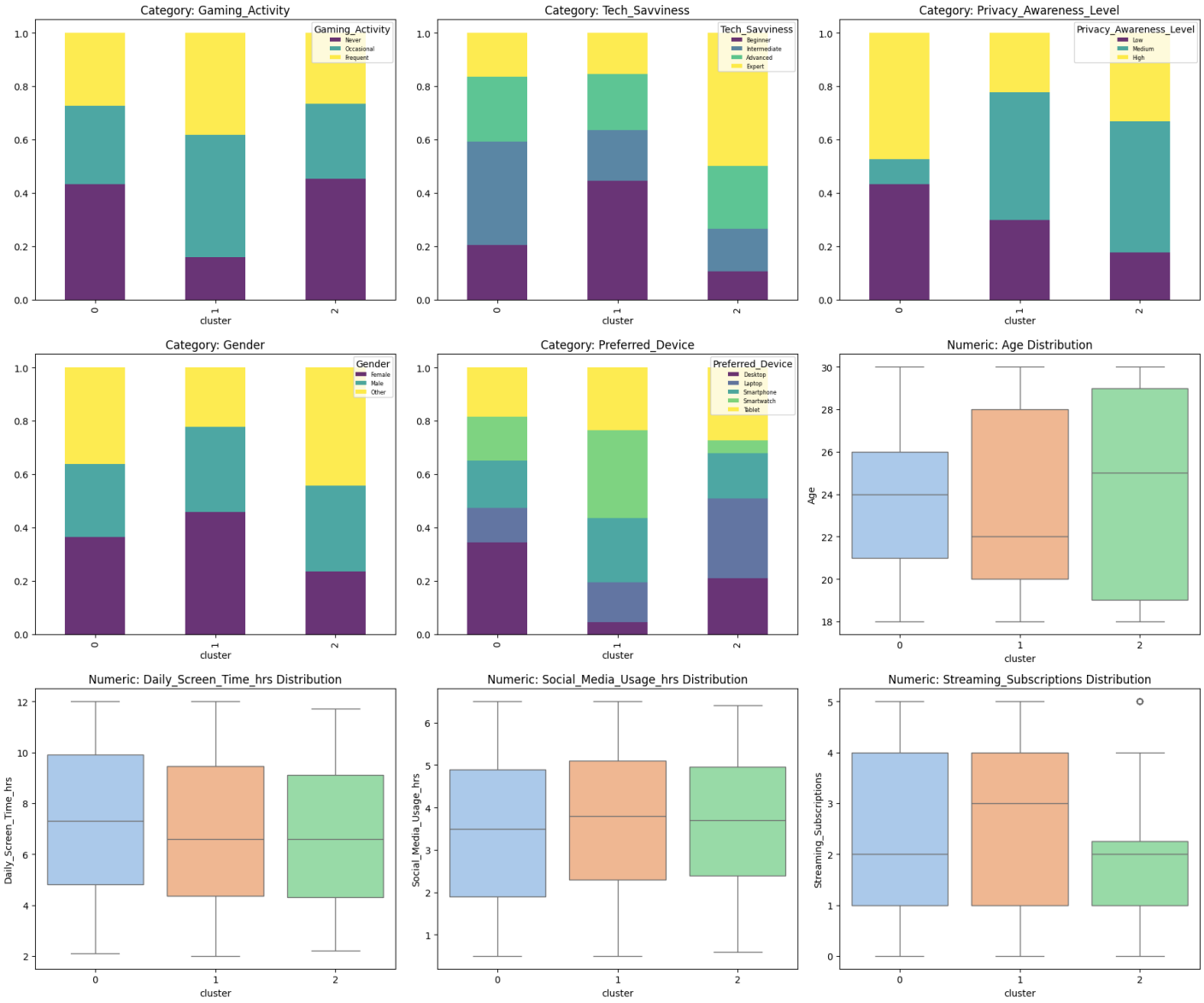


Figure (6) Distribution of Categorical and Numerical Variables in Each Cluster

Linking the Cluster's Profile to Dimension's coordinates

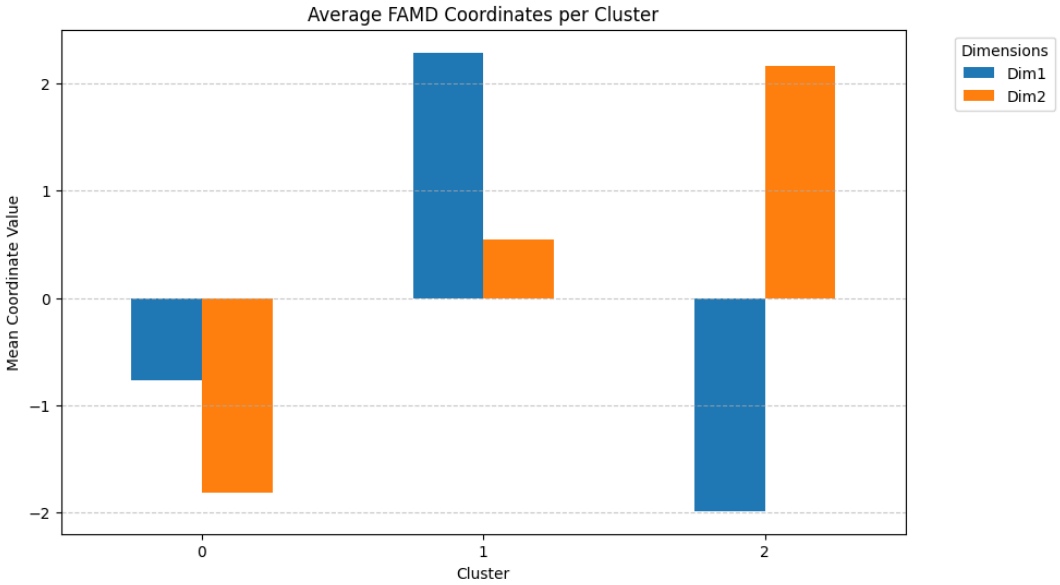


Figure (7) Average FAMD Coordinates per Cluster

Cluster 0 Vs Dimensions

This cluster has a significant and equal proportion of female and 'other' gender type. Females pull toward to the right end of Dim 1 and 'Other' gender moves toward left side of Dim1. Despite weak correlation of 'age', the concentrated middle-aged group will be situated in the bottom-right of the map. While this cluster moves toward negative ends of Dim1 and Dim2 regarding the highest 'daily screen time', they are more likely to engage in 'streaming' services (right side of Dim1) rather than 'social media' (positive on both Dim1 and Dim2). Having a large proportion of 'non-gamers', they are being pulled into the negative space of Dim1 and Dim2. Considering the small proportion of beginner level and moderate proportion of intermediate level 'tech savvy', this cluster will sit near the origin (0,0). Regarding 'privacy awareness', this cluster is defined by the polarity of low and high levels, which is reflected in its position within the negative coordinates of Dim1 and Dim2. Given its 'desktop' preference, the cluster is also pulled toward negative poles of Dim1 and Dim2.

Cluster 0 Persona: This seems to align with the profile of about 20-to-26-years old who have high screen time, probably with steady income to pay for streaming services. They have a solid grasp of technology for daily life and may play a casual game once in a while. They either do not care the privacy at all or are highly aware, with no middle ground. (Non-binary gender may prefer gadgets designed with user privacy as their primary value). While they are generally desktop traditionalists or stationary surfers, females may like smartphones to engage in streaming.

Cluster 1 Vs Dimensions

This cluster has the highest female proportion and pulls toward to the right end of Dim 1 and lower side of Dim2. Not being a primary driver of the differences, this youngest group moves slightly against the bottom-right of the map. With the highest subscription in streaming services and the highest usage of social media, they lean toward the right side Dim1. Having the highest proportion of 'gamers', they are pushed further into the positive space of Dim1 and Dim2. The largest proportion of beginner level 'tech savvy' moves toward the right-bottom of the map. This cluster includes users with low-to-mid-level privacy concerns, which reflects the positive coordinates of Dim1 and Dim2. With a focus on portable devices, the cluster is the top user of smartwatch and pulls toward the positive pole of Dim1 and the negative pole Dim2.

Cluster 1 Persona: The youngest female dominant tech-forward user group with median age of 22. With the highest digital consumption (streaming & social media), they are actively integrating gaming into their lifestyle and likely exploring tech as a primary hobby. The portable devices (smartwatch, smartphone, or tablet) will be their preference with active beginner level tech savvy and average privacy concern.

Cluster 2 Vs Dimensions

The cluster includes the largest proportion of 'other' gender type and pulls toward to the top-left of the map. With a very wide range, the oldest demographic (median ~25) is situated in the bottom-right space of the map. Interestingly, the cluster will be pulled toward the negative sides of Dim1 and Dim2 as it has the lowest values in 'daily screen time', 'social media usage' and 'streaming subscription'. Having the highest proportion of 'non-gamers', they are being pulled into the negative space of Dim1 and Dim2. The highest proportion of 'tech expert' are pushed toward the negative side of Dim 1 and the positive side of Dim2. Regarding 'privacy awareness', this cluster is defined by the large proportion of privacy-conscious individuals, and it will be positioned in the positive territory of Dim1 and Dim2. Being 'laptop' as a preference for a significant proportion of the cluster, it is also pulled toward negative pole of Dim1 and the positive pole of Dim2.

Cluster 2 Persona: This corresponds to the 'Professional' profile of an individual approximately 25 years of age who identifies as non-binary or gender-diverse. These individuals with low-digital footprints likely use technology strictly for productivity or work rather than for gaming, streaming or social media. They are also highly tech-literate. Privacy-focused laptops or tablets will be their favorites for the privacy-conscious minimalists.